

NIDS Session Report

Start Time: 1750243082.5906634

Stop Time: 2025-06-18T16:09:39.994925

Packets Processed: 302

Anomalies Detected: 19

Packet Rate: 0 packets/sec

Protocol Stats:

total_packets: 302

tcp_packets: 221

udp_packets: 72

icmp_packets: 0

other_packets: 9

bytes_captured: 66073

start_time: 1750243082.5883868

uptime: 97.40652632713318

packets_per_second: 3.100408272293312

bytes_per_second: 678.3221052160133

Total Alerts: 5

Alerts:

Type: ml_anomaly | Time: 2025-06-18T16:08:07.630864

Details: {'anomaly_score': -7.245024138727217, 'features': {'packet_count': 16, 'total_bytes': 946, 'avg_packet_length': np.float64(59.125), 'std_packet_length': np.float64(4.525966747557918), 'min_packet_length': 42, 'max_packet_length': 64, 'avg_payload_size': np.float64(1.75), 'std_payload_size': np.float64(0.6614378277661477), 'total_payload_size': 28, 'tcp_count': 14, 'udp_count': 0, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 1, 'tcp_ratio': 1.0, 'udp_ratio': 0.0, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 1, 'unique_dst_ports': 14, 'total_ports_accessed': 15, 'common_port_ratio': 0.0, 'high_port_ratio': 1.0, 'avg_dst_port': np.float64(51456.78571428572), 'std_dst_port': np.float64(7.551523700822347), 'min_dst_port': 51446, 'max_dst_port': 51467, 'unique_connections': 14, 'total_connection_attempts': 14, 'avg_packets_per_connection': 1.0, 'connection_diversity': 0.875, 'max_packets_per_connection': 1, 'unique_src_ips': 14, 'unique_dst_ips': 1, 'src_ip_diversity': 0.875, 'dst_ip_diversity': 0.0625, 'private_src_ip_ratio': 0.0, 'private_dst_ip_ratio': 1.0, 'packet_rate': 6.434925273153346, 'inter_arrival_time_mean': np.float64(0.1657620906829834), 'inter_arrival_time_std':

np.float64(0.23400472516283943), 'burst_count': 10, 'burst_ratio': 0.6666666666666666, 'payload_size_mean': np.float64(1.75), 'payload_size_std': np.float64(0.6614378277661477), 'zero_payload_ratio': 0.125, 'large_payload_ratio': 0.0, 'payload_size_entropy': np.float64(0.5435644431995964), 'syn_count': 14, 'fin_count': 0, 'rst_count': 0, 'psh_count': 0, 'ack_count': 14, 'urg_count': 0, 'syn_ratio': 1.0, 'fin_ratio': 0.0, 'rst_ratio': 0.0, 'psh_ratio': 0.0, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:08:07.630848', 'packet_count': 16, 'protocol_distribution': {'tcp': 14, 'udp': 0, 'icmp': 0}, 'connection_info': {'unique_connections': 14, 'unique_src_ips': 14, 'unique_dst_ips': 1}}

Type: port_scan | Time: 2025-06-18T16:08:14.738405

Details: {'src_ip': '10.0.2.15', 'dst_ip': '10.0.2.3', 'dst_port': 53, 'protocol': 17}

Type: ddos_indicator | Time: 2025-06-18T16:08:17.130769

Details: {'src_ip': '3.234.165.166', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Type: ml_anomaly | Time: 2025-06-18T16:09:07.823333

Details: {'anomaly_score': -7.226891511707684, 'features': {'packet_count': 100, 'total_bytes': 28401, 'avg_packet_length': np.float64(284.01), 'std_packet_length': np.float64(437.56648626237364), 'min_packet_length': 54, 'max_packet_length': 1429, 'avg_payload_size': np.float64(233.24), 'std_payload_size': np.float64(441.5864608431739), 'total_payload_size': 23324, 'tcp_count': 56, 'udp_count': 42, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.5714285714285714, 'udp_ratio': 0.42857142857142855, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 7, 'unique_dst_ports': 23, 'total_ports_accessed': 23, 'common_port_ratio': 0.2653061224489796, 'high_port_ratio': 0.7346938775510204, 'avg_dst_port': np.float64(38136.08163265306), 'std_dst_port': np.float64(22664.664409817822), 'min_dst_port': 443, 'max_dst_port': 54000, 'unique_connections': 24, 'total_connection_attempts': 98, 'avg_packets_per_connection': 4.083333333333333, 'connection_diversity': 0.24, 'max_packets_per_connection': 13, 'unique_src_ips': 21, 'unique_dst_ips': 5, 'src_ip_diversity': 0.21, 'dst_ip_diversity': 0.05, 'private_src_ip_ratio': 0.047619047619047616, 'private_dst_ip_ratio': 0.2, 'packet_rate': 2.596313478552065, 'inter_arrival_time_mean': np.float64(0.3890520225871693), 'inter_arrival_time_std': np.float64(0.9239629062857324), 'burst_count': 54, 'burst_ratio': 0.5454545454545454, 'payload_size_mean': np.float64(233.24), 'payload_size_std': np.float64(441.5864608431739), 'zero_payload_ratio': 0.03, 'large_payload_ratio': 0.14, 'payload_size_entropy': np.float64(1.2224015904695353), 'syn_count': 35, 'fin_count': 0, 'rst_count': 9, 'psh_count': 3, 'ack_count': 56, 'urg_count': 0, 'syn_ratio': 0.625, 'fin_ratio': 0.0, 'rst_ratio': 0.16071428571428573, 'psh_ratio': 0.05357142857142857, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:09:07.823319', 'packet_count': 100, 'protocol_distribution': {'tcp': 56, 'udp': 42, 'icmp': 0}, 'connection_info': {'unique_connections': 24, 'unique_src_ips': 21, 'unique_dst_ips': 5}}

Type: port_scan | Time: 2025-06-18T16:09:14.746651

Details: {'src_ip': '10.0.2.15', 'dst_ip': '23.21.130.158', 'dst_port': 443, 'protocol': 6}